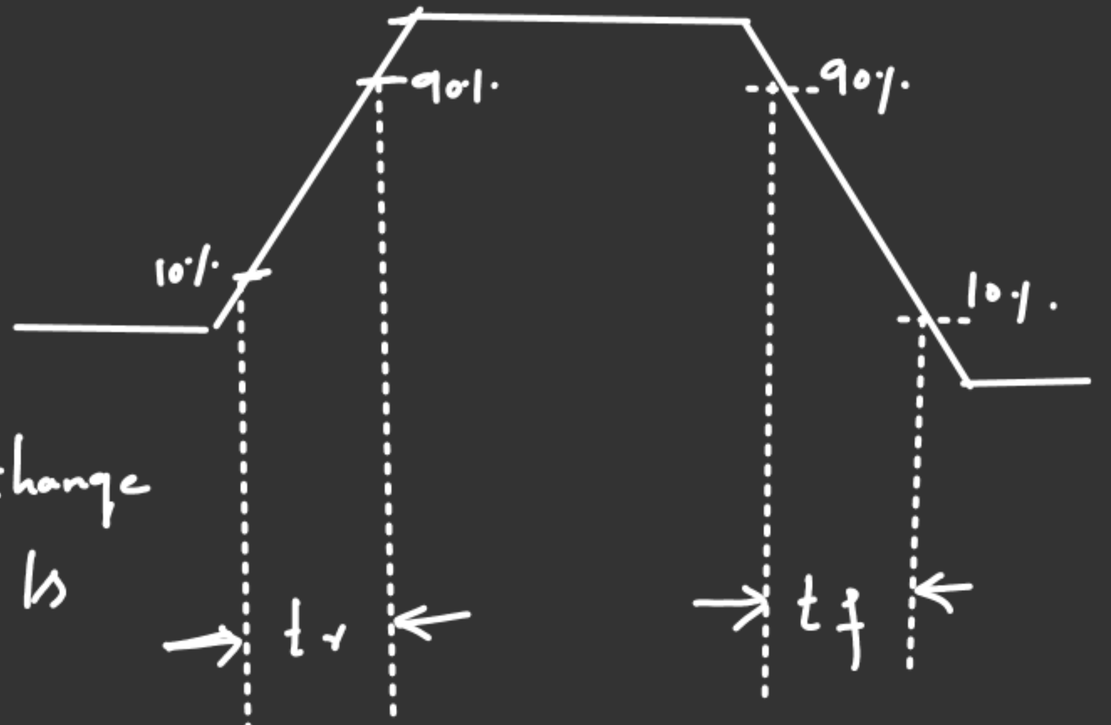
Transition.



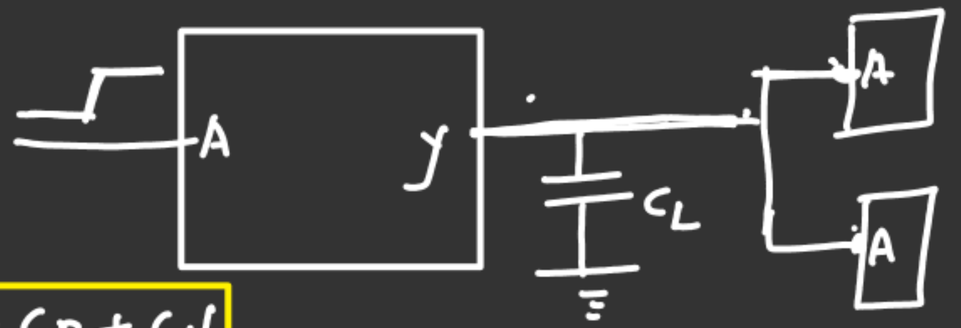


Transition time: Time taken by the signal to change from 10% of max% Value to 90% max% Value is called rise transition

Time taken by signal to change from 90% of max% Value to 10% of max% Value is called as fall transition.

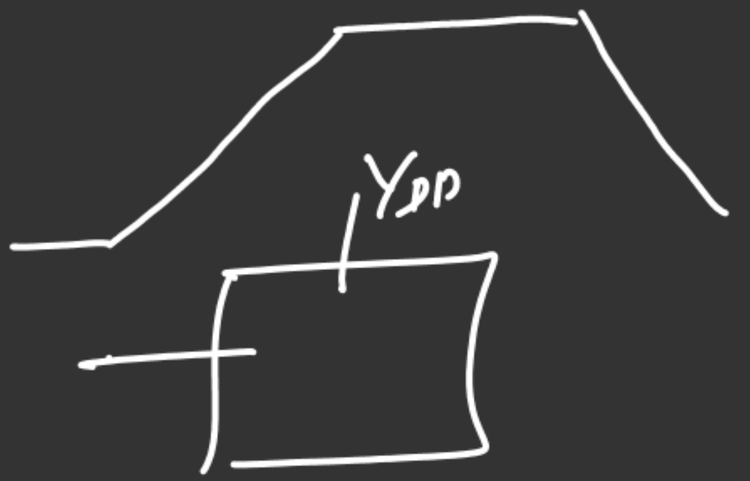
o/p transition Analysis.

Parameters	Parameter value	transition value / Cell value
Load Capacitance C_L	Increases	Increases.
Input transition	Increases	Increases.
(W/L) →	Increases, id increases	decreases.
V_{DD} →	Increases, id increases	decreases
μ_n →	Increases, id increases	decreases. } temperature
V_{th} →	Increases, id decreases	Increases } temperature



$C_L = C_p + C_N$

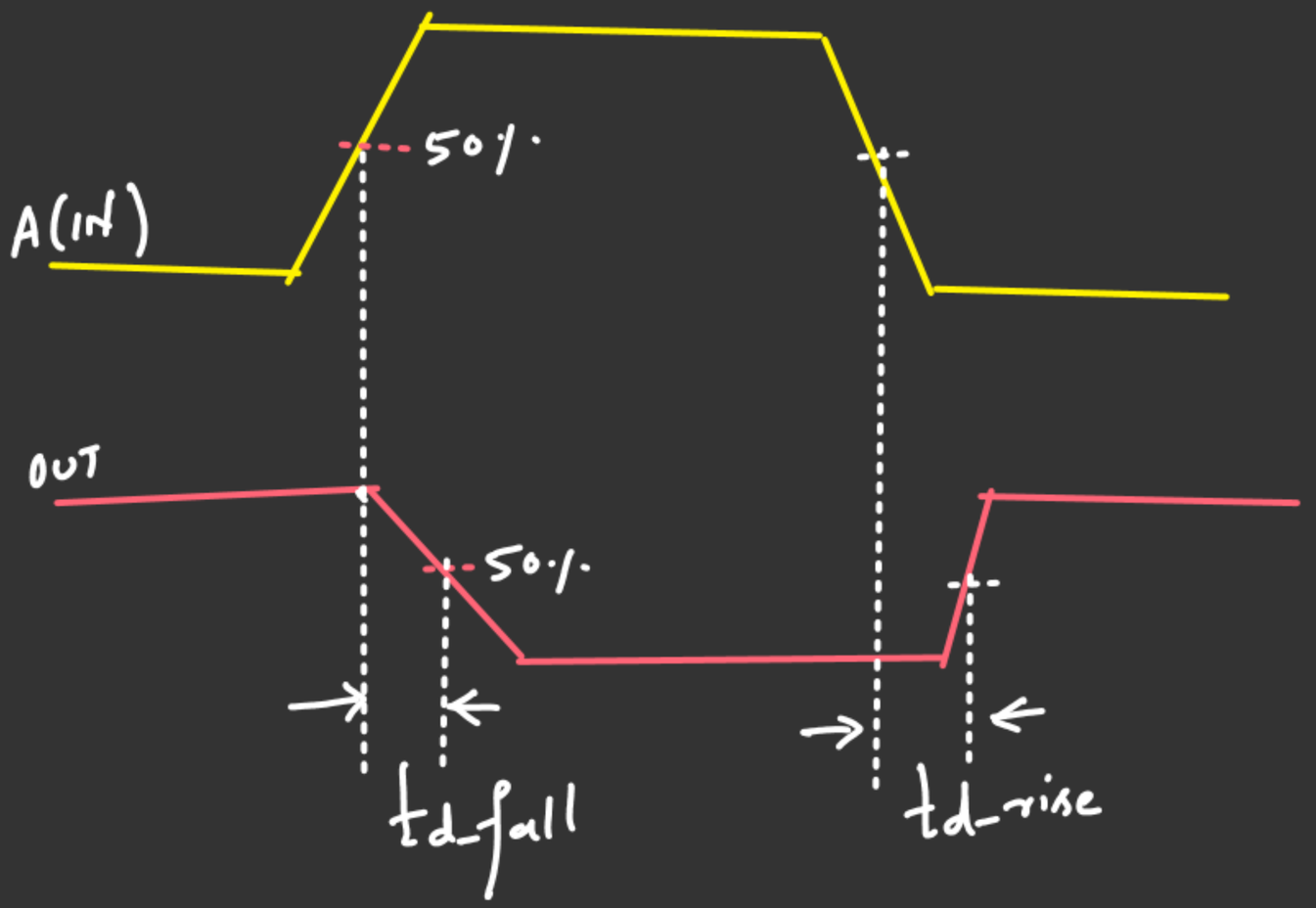
$$\uparrow i_d = \frac{\mu_n C_{ox}}{2} (W/L) (V_{gs} - V_{th})^2 \uparrow$$



Cell_delay:



Time difference b/w 50% of i/p tranition to 50% of output tranition is called as cell delay.



PARAMETERS		Drain Current (Id)	Output Transition And Cell delay
Load Capacitance	Increase	Increases	Increases
	Decrease	Decreases	Decreases
Input Transition	Increase	Increases	Increases
	Decrease	Decreases	Decreases
(W/L) ratio	Increase	Increase	Decrease
	Decrease	Decrease	Increase
Vdd	Increase	Increases	Decreases
	Decrease	Decreases	Increases
Mobility μ_n / μ_p	Increase	Increases	Decreases
	Decrease	Decreases	Increase
Threshold Voltage (Vth)	Increase	Decreases	Increases
	Decrease	Increases	Decreases

(*) More (W/L) ratio → Less cell delay.

Less V_{DD} → More cell delay

More C_L → More cell delay

More Net length → More cell delay.

More fanout → More cell delay.

Less i/p trans → Less cell delay.

V_{th} Less → Less cell delay

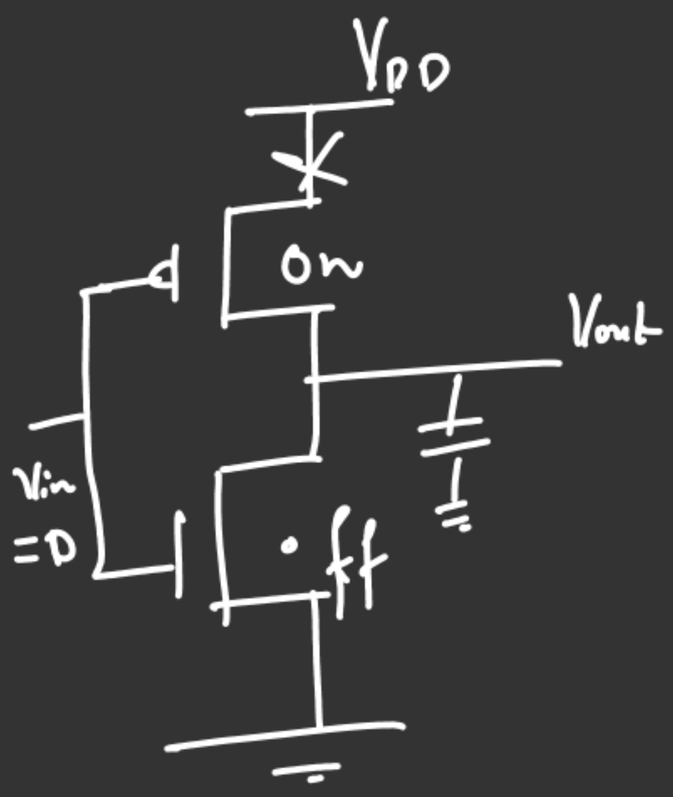
Power:

- Static → Leakage power
- Dynamic → Short ckt and Switching power.

Static → power dissipated when i/p of cell is not switching.

Dynamic → power dissipated when i/p of cells are switching.

Static Power:



↳ Power dissipated due to Leakage Current

↳ Current Conducted even when MOSFET is in off state.

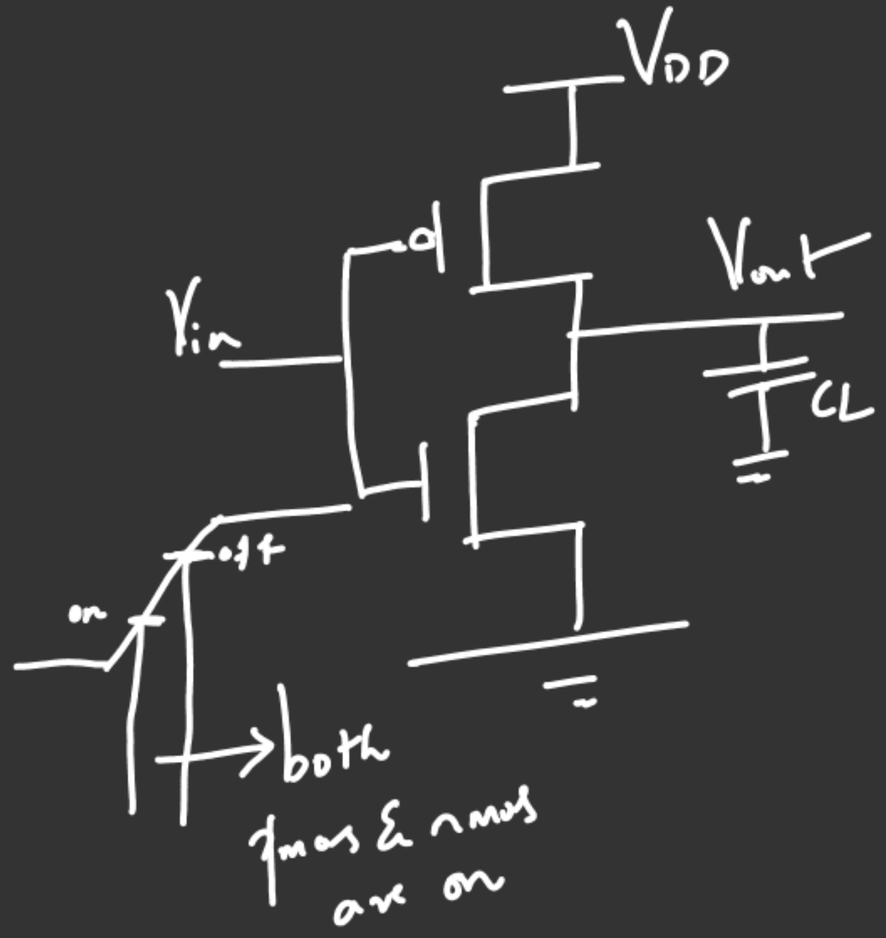
↳ V_{th} → More, Leakage Current is Less

V_{th} → Less, More Leakage Current

↳ V_{DD} (↑) → ↑ Leakage

V_{DD} (↓) → ↓ Leakage

Short circuit power.



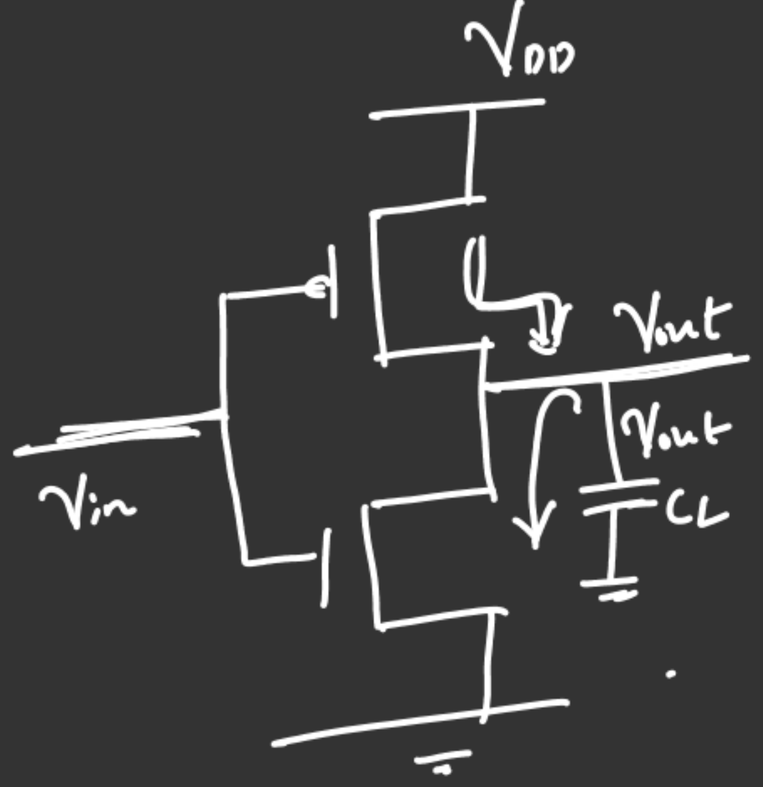
Power dissipated due direct current flow b/w V_{DD} & ground when i/p is switching.

↳ V_{DD} ↑ → short ckt power ↑

↳ i/p transition → ↑ short ckt power

↳ frequency of switching ↑ → short ckt power ↑

Switching Power



(2/3)

Power dissipated due to charging & discharging of Load Capacitance

$$S.P = \frac{1}{2} \cdot \alpha \cdot f \cdot C_L \cdot V_{dd}^2$$

(1/2)

- C_L → Load Capacitance
- f → frequency of clock
- α → Switching activity (2/6)

Different Cells with Same functionality

drive strength

INVX1 → (W/L) Less → More cell delay → Less area

INVX2 → (W/L) = 2 × (W/L)$_{X1}$ → Less cell delay → More area

INVX4 → (W/L) = 4 × (W/L)$_{X1}$ → Very less cell delay → More area than X_2 cell

INVX8 → (W/L) = 8 × (W/L)$_{X1}$ → Least cell delay → huge area

1

Cells with different Vth.

INV-HVT → More cell delay → Less Leakage power

INV-RVT → Less cell delay → More Leakage power

INV-LVT → Very less cell delay → high Leakage power

INV-ULVT → Least cell delay → Very high leakage power